

Wearable, skin-like heart monitor

Prof. Zhenan Bao and his team at the Stanford University have developed a tiny, paper-thin, extremely accurate sensor to monitor a person's pulse. According to the university's press release, the device is made by combining layers of flexible materials into pressure sensors, and is thinner than a dollar bill and no wider than a postage stamp. With this, it would be possible for doctors to continuously track heart health and detect cardiovascular problems accurately and quickly.

Inside: When you place your thumb on your wrist, you feel a constant beat—your pulse. Each of these beats is actually made up of two distinct peaks, which we cannot make out. The first is from your heart pumping out blood, and the second is from the reflecting wave sent back from the lower body to the artery system. The first is long and the second short; the relative sizes of these two peaks can be used by doctors to measure various factors of heart health, such as the stiffness of the artery.

The small and flexible device developed by Bao's team is sensitive enough to monitor both the peaks. The device, which is to be taped on using a simple Band-Aid, has a thin middle layer of rubber covered with tiny pyramid bumps. Each mould-made pyramid is only a few microns in size.

When pressure is put on the device, the pyramids deform slightly, changing the size of the gap between the two halves of the device. This change in separation causes a measurable change in the electromagnetic field and the current flow in the device. The more the pressure placed on the monitor, the more the pyramids deform and the larger the change in the electromagnetic field.

When the sensor is placed on someone's wrist, it can measure that person's pulse wave as it reverberates through the body. Apart from the main peak and the recoil, the highly-sensitive device also detects another tiny bump in the tail of the graph, which could also be useful information.

Web: <http://news.stanford.edu/news/2013/may/skin-heart-monitor-051413.html>



Unmanned storm chasers

Storm chasing is a perilous job. As an alternative to manned aircrafts chasing storms to gather meteorological data, three teams of students led by Professor Jamey Jacob at the Oklahoma State University (OSU) have developed three concept storm-penetrating air vehicles (SPAVs) capable of penetrating thunderstorms right up to the super-cells that generate the tornado. If productised, these drones could collect data that would help storm trackers and researchers to understand tornadoes better.

Inside: The SPAVs meet several preset standards. They are designed to be able to take-off from a traditional road even in bad conditions like 35km-per-hour winds with gusts up to 45 km per hour, fit inside a standard flatbed trailer, and fly for at least four hours at 1.5 km (5000 feet) without refuelling. They have also been designed to carry and drop one or more dropsondes (cylinders full of sensors) into the tornado for collecting data about temperature, humidity, wind speed and wind direction. Older forms of data collection use the Doppler radar, which provides information about moisture levels but not temperature gradients, pressure levels, etc. However, sensors in dropsondes should hopefully be able to collect more thermodynamic and meteorological information that could possibly reveal more about the nature of tornadoes.

OSU is negotiating with its partners, including the University of Colorado at Boulder to further these designs into full-fledged development.

Web: <https://news.okstate.edu/press-releases/2186-osu-students-design-tornado-chasing-uavs>

